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Anomaly detection based on fuzzy neighborhood rough sets

Yuan Yuan [a,b], Sihan Wang [a], Hongmei Chen [c], Chuan Luo [a], Zhong Yuan [a,*]

[a] College of Computer Science, Sichuan University, Chengdu 610065, China
[b] School of Software Technology, Zhejiang University, Ningbo, 315048, China
[c] School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu 611756, China

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ABSTRACT

Anomaly detection is a significant data mining task with great application potential. However, the presence of uncertain information, such as noise and fuzziness, in real data usually poses a great challenge to existing anomaly detection methods. As an important technique of granular computing theory, fuzzy neighborhood rough sets can effectively handle uncertain data. It has been successfully applied to data pre-processing tasks such as attribute reduction or feature selection. In response to the shortcomings of existing anomaly detection methods that cannot effectively handle uncertain information such as noise and fuzziness, this paper proposes an unsupervised anomaly detection algorithm based on a novel fuzzy neighborhood rough sets model. At first, parameterized fuzzy relations are introduced to inscribe fuzzy neighborhood information granules. By establishing the relationship between different granules, the fuzzy neighborhood lower and upper approximations are defined to construct a fuzzy neighborhood rough set model for decision-free information systems. The lower approximation ratio is further proposed and used to construct the anomaly score, which characterizes the anomaly degree of the objects by aggregating the granule intensity of anomalies with corresponding weights. Finally, an unsupervised anomaly detection algorithm is designed for heterogeneous data based on the proposed fuzzy neighborhood rough set model. The results of extensive experiments illustrate that the proposed algorithm outperforms the other ten comparison algorithms and can effectively handle uncertain information. The code is publicly available online at https://github.com/yuyuan97/ADFNR.

1. Introduction

Anomalies, also known as outliers, are data points whose attributes or behaviors are different from the observed objects. The purpose of anomaly detection is to identify the object points that are quite different from normal objects. Initially, it is often used as a pre-processing operation before data analysis to reduce the impact of outliers on the results of analysis. Nowadays, anomaly detection has emerged as a vital data-mining task with the aim of exploring the latent value contained in anomalies. The importance and potential harm of anomalies create a wide range of needs and great application potential for anomaly detection, such as intrusion detection [1], fraud detection [2], power monitoring [3] and so on, involving military, financial, medical, ecological, security, and

* Corresponding author.
E-mail address: yuanzhong@scu.edu.cn (Z. Yuan).

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other fields. The particularity and timeliness of the anomaly detection tasks put forward higher requirements for the speed and accuracy of the algorithm.

According to different research methods, anomaly detection algorithms can be divided into five categories: depth-based [4], distribution-based [5], clustering-based [6], distance-based [7], and density-based [8]. In recent years, with the deepening of Granular Computing (GrC) theory, anomaly detection models based on GrC have been gradually proposed. GrC, first proposed by Zadeh, has been proven to be an important method for dealing with uncertain and fuzzy data. A granule is defined as a collection of points or objects clustered together due to some rule such as indistinguishability or similarity [9]. As one of the main GrC models, the rough set model is a mathematical tool that effectively describes incompleteness and uncertainty of information, and has been proven to be effective in outlier detection [10]. However, the classical rough set theory is based on equivalence classes, so numerical data need to be discretized before further analysis, which may lead to information loss and speed reduction.

To eliminate the impact of discretization, scholars have proposed a variety of extended forms of rough sets. For example, fuzzy sets and rough sets are two extensions of elementary set theory. Combining the advantages of fuzzy set and rough set models, Dubois and Prade [11] first proposed the fuzzy rough set model by using fuzzy equivalence relations instead of equivalence relations. It has been utilized in feature extraction, clustering, and anomaly detection [12]. Lin et al. [13] proposed the concept of neighborhood rough sets, which can be directly applied to numerical and heterogeneous data by using neighborhood similar classes with thresholds to avoid the discretization process. However, it cannot effectively handle the fuzziness of instances [14]. Therefore, Wang et al. [14] proposed a fuzzy neighborhood rough set (FNRS) model and designed a greedy feature subset selection algorithm. Based on FNRS, some generalizations were proposed and successfully applied to feature subset selection [15]. FNRS theory combines the advantages of fuzzy rough sets and neighborhood rough sets and provides an effective tool to deal with uncertain data.

While FNRSs have demonstrated value in feature selection, their potential for anomaly detection remains underexplored. Critically, existing FNRS models rely on supervised decision attributes, inherently limiting their applicability to unsupervised tasks like anomaly detection. This limitation motivates our work: (1) We identify the structural incompatibility between conventional FNRS frameworks and unsupervised anomaly detection requirements as a key limitation; (2) To address this, we propose a generalized FNRS model that eliminates dependency on decision attributes through neighborhood granulation and uncertainty quantification; (3) Building upon this theoretical extension, we develop the first FNRS-based anomaly detection algorithm specifically designed for heterogeneous data. Our approach not only expands the mathematical foundation of FNRS but also demonstrates its practical viability in handling uncertain, high-dimensional data patterns that challenge conventional methods as well as broadens its application domains. Specifically, in this paper, we first use parameterized fuzzy relations to inscribe fuzzy neighborhood information granules. The upper and lower approximations are then introduced to construct an FNRS model for decision-free information systems. Further, based on the fuzzy neighborhood lower approximation ratio, the Granularity Intensity of Anomalies (GIA) is defined to describe the degree of anomalies of fuzzy neighborhood granules. In order to measure the anomaly degree of each object more comprehensively, the GIA is further combined with the corresponding weights to propose the fuzzy neighborhood granular-based anomaly score. Finally, an Anomaly Detection based on the Fuzzy Neighborhood Rough sets (ADFNR) algorithm is designed for heterogeneous data. Experimental results on publicly available datasets show that the algorithm has good performance in comparison with state-of-the-art anomaly detection methods.

The remainder of this article is organized as follows. In Section 2, we review some relevant research on anomaly detection based on rough sets and extended models. Section 3 presents some preliminaries of FNRS theory that are relevant to this paper. In Section 4, a generalized FNRS model is introduced to be used in the decision-free information system. In Section 5, we design an FNRS-based anomaly detection algorithm. Then, the experimental details and analysis of the experimental results are shown in Section 6. Finally, Section 7 summarizes the core contributions of this work and proposes novel research directions to address remaining challenges.

2. Related works

Rough set theory has been proven to be effective in anomaly detection, especially for nominal data. For example, Jiang et al. [16] applied rough set theory to outlier detection by proposing sequence-based and distance-based outlier detections. Albanese et al. [17] extended outlier detection to spatiotemporal data by using the rough set approximations to define outliers. Jiang and Chen [10] introduced a new outlier detection method based on GrC and rough set theory. However, rough set theory-based methods didn't perform well on numerical data, as important information is lost when it is discretized.

As a significant extension of rough set theory, fuzzy rough set theory effectively addresses the limitation of classical rough sets in handling numerical data. Anomaly detection algorithms based on the fuzzy rough set theory have been proposed over the years. For example, to address the limitations that existing anomaly detection methods based on fuzzy rough sets cannot be applied to mixed data, Yuan et al. [18] proposed a new fuzzy rough set model using a mixed fuzzy similarity relation and designed a specific Fuzzy Rough Granules-based Outlier Detection (FRGOD) algorithm. Yuan et al. [12] also constructed an outlier detection algorithm based on fuzzy information entropy to effectively handle hybrid data. Wang et al. [19] constructed distance-based fuzzy rough entropy for characterizing the samples, and finally weighted and summarized the fuzzy rough relative entropy of two attribute sequences for anomaly detection. Yuan et al. [20] proposed a multi-fuzzy granules anomaly detection method by introducing the idea of multi-granularity to the fuzzy rough computing models. Although the fuzzy rough set theory has been verified to be effective in outlier detection, it still has the limitation of being sensitive to noise.

Neighborhood rough sets also play an important role in outlier detection. By introducing neighborhood information to the traditional distance metric, Chen et al. [21] proposed a neighborhood-based anomaly detection algorithm. Goh et al. [22] presented an anomaly detection method by using neighborhood rough sets with correctness matching, which can effectively handle mixed datasets and nominal datasets. Wang et al. [23] constructed a weighted neighborhood information network to represent the mixed-value attribute dataset by considering the neighborhood similarities and relations among samples, and finally used Markov random wandering method to detect outliers. Li et al. [24] detected outliers by using information entropy to measure the underlying mechanism of uncertainty based on neighborhood rough sets, which can handle real-valued data and preserve the semantics of datasets. Yuan et al. [25] introduced neighborhood information entropy and constructed the corresponding anomaly factor and anomaly detection algorithm, effectively improving the applicability of outlier detection in hybrid data.

The FNRS model was first proposed by Wang et al. [14] and applied to feature subset selection, opening the way to select feature subsets using FNRS. On this basis, Sun et al. [26] combined FNRS with multigranulation rough sets (MRS) to construct fuzzy neighborhood multigranulation rough sets (FNMRS) in neighborhood systems and then presented an FNMRS-based feature selection approach by using fuzzy neighborhood pessimistic multigranulation entropy. In classification tasks on multilabel data with missing labels, Sun et al. [27] proposed a multilabel fuzzy neighborhood rough sets (MFNRS) model to improve the robust performance, which makes a combination of the multilabel neighborhood rough sets (MNRS) with FNRS. Suo et al. [28] proposed a variable precision model based on neighborhood rough sets, then defined a new theory of fuzzy rule acquisition and decision-making, and finally performed fault diagnosis of the geosynchronous orbit satellite power system. By FNRS, the fuzzy-neighborhood relative decision entropy was proposed and applied to construct a heuristic feature selection algorithm by Zhang et al. [15]. However, the research on the effectiveness of FNRS for anomaly detection has not been reported.

3. Preliminaries

In this section, we review some important definitions of the FNRS model [14] that are relevant to this paper.

The decision table in an FNRS model can be defined by $\langle O, A, D\rangle$, where $O=\left\{o_1, o_2, \ldots, o_n\right\}$ represents a set of objects; A is a set of conditional attributes; and D is a set of decision attributes. For any $B \subseteq A$ and $a \in B$, a fuzzy similarity relation induced by B can be denoted by $R_B=\cap_{a \in B} R_a$ if R_B satisfies the reflexivity and symmetry: (1) $R_B(o, o)=1$, for any $o \in O$; (2) $R_B(o, q)=R_B(q, o)$, for any $o, q \in O$. R_B can also be denoted as a matrix in which each element $R_B(o, q)$ is the similarity between o and q. For any $B_1, B_2 \subseteq A$, and any $o_i, o_j \in O$, some widely used operations of fuzzy relations are defined as follows.

1) $R_{B_1}(o_i, o_j) \leq R_{B_2}(o_i, o_j) \Rightarrow R_{B_1} \subseteq R_{B_2}$;
2) $(R_{B_1} \cap R_{B_2})(o_i, o_j)=\min\left\{R_{B_1}(o_i, o_j), R_{B_2}(o_i, o_j)\right\}$;
3) $(R_{B_1} \cup R_{B_2})(o_i, o_j)=\max\left\{R_{B_1}(o_i, o_j), R_{B_2}(o_i, o_j)\right\}$.

Definition 1. Given a data table $\langle O, A\rangle$, $B \subseteq A$, and the radius of neighborhood $\varepsilon \in (0,1]$, for any $o, q \in O$, the fuzzy neighborhood information granule of o with reference to B is denoted by

$$[o]_B^{\varepsilon}(q)=\begin{cases}0, & R_B(o, q)<\varepsilon; \\ R_B(o, q), & R_B(o, q) \geq \varepsilon.\end{cases} \tag{1}$$

Let $[o_i]_B^{\varepsilon}=\left\{[o_i]_B^{\varepsilon}(o_1), [o_i]_B^{\varepsilon}(o_2), \ldots, [o_i]_B^{\varepsilon}(o_n)\right\}$, for any $o_i \in O$. The cardinality of $[o_i]_B^{\varepsilon}$ is $\left|[o_i]_B^{\varepsilon}\right|=\sum_{j=1}^n [o_i]_B^{\varepsilon}(o_j)$. Given $B, B_1, B_2 \subseteq A$, for any $o \in O$, the following properties are obtainable.

1) If $B_1 \subseteq B_2$, then $R_{B_1} \supseteq R_{B_2}$.
2) If $B_1 \subseteq B_2$, then $[o]_{B_1}^{\varepsilon} \supseteq [o]_{B_2}^{\varepsilon}$.
3) If $\varepsilon_1 \leq \varepsilon_2$, then $[o]_B^{\varepsilon_1} \supseteq [o]_B^{\varepsilon_2}$.

Definition 2. Given a decision table $\langle O, A, D\rangle$, the fuzzy decision classification $O/D=\left\{D_1, D_2, \ldots, D_l\right\}$, $B \subseteq A$, for any $o \in O$, fuzzy neighborhood lower and upper approximations of D_j with respect to B are respectively denoted by

$$\underline{N}_B^{\varepsilon}\left(D_j\right)=\left\{o \in O \mid [o]_B^{\varepsilon} \subseteq D_j\right\}, \tag{2}$$

$$\overline{N}_B^{\varepsilon}\left(D_j\right)=\left\{o \in O \mid [o]_B^{\varepsilon} \cap D_j \neq \emptyset\right\}. \tag{3}$$

The relationship between the fuzzy neighborhood granule of an object and the decision class determines whether the object belongs to the upper or lower approximation. $\underline{N}_B^{\varepsilon}\left(D_j\right)$ consists of objects that completely belong to D_j while $\overline{N}_B^{\varepsilon}\left(D_j\right)$ consists of objects that probably belong to D_j.

Definition 3. Given a decision table $\langle O, A, D\rangle$, $O/D=\left\{D_1, D_2, \ldots, D_l\right\}$, $B \subseteq A$, for any $o \in O$, fuzzy neighborhood positive region and dependency degree of D with respect to B are defined as

$$POS_B^{\varepsilon}(D)=\bigcup_{j=1}^{l} \underline{N}_B^{\varepsilon}\left(D_j\right), \tag{4}$$

$$\gamma_B^{\varepsilon}(D) = \frac{\left|POS_B^{\varepsilon}(D)\right|}{|O|}, \tag{5}$$

where $|\cdot|$ stands for the cardinality of a set.

The dependency degree reflects the correlation between decision attributes and conditional attributes. It is determined by the size of the positive region and the number of objects. Both the positive region and dependency degree are monotonic on the attribute subset and the neighborhood radius [14]. However, as we can see from the above definitions, the computation of the dependency degree and the positive region relies on the information of the decision attributes. This leads to their incapability of handling uncertain data in unsupervised tasks which do not contain decision attributes and also limits the application fields of the FNRS theory. In view of the above discussion, it becomes a focus of our work to develop a generalized FNRS model that can be used in decision-free information systems. This will contribute to the extension of FNRS theory and its further application to anomaly detection.

4. Generalized FNRS model

The FNRS model combines the advantages of fuzzy rough sets and neighborhood rough sets well, but it is built on a decision system, which limits its application scope. To enhance the applicability of FNRS theory, we propose a generalized FNRS model: an FNRS model for decision-free information systems.

Let a quintuple $\langle O, A, V, f, \varepsilon\rangle$ represent a decision-free fuzzy neighborhood information system (FNIS), where $O = \left\{o_1, o_2, \ldots, o_n\right\}$ is a set of objects and $A = \left\{a_1, a_2, \ldots, a_m\right\}$ is a set of attributes. Both sets are non-empty finite. Let V_a represent the value domain of a single attribute, then V can be expressed as $V = \bigcup_{a\in A} V_a$. f is the map function $f : O \times A \to V$ and $f(o, a)$ is the attribute value of object o taken on attribute a. The neighborhood radius $\varepsilon \in [0, 1]$ is an essential parameter. It exerts a profound influence on the model's behavior, shaping the nature of fuzzy neighborhood granule structures and, consequently, the overall performance of the model in handling decision-free information systems. A smaller value of the neighborhood radius ε restricts the scope of the fuzzy neighborhood granules. This leads to a more refined and detailed view of the data, as the granules are centered around individual objects and capture local-level characteristics more precisely. For instance, in datasets where the data points exhibit distinct local patterns or fine-grained differences, a small ε enables the model to distinguish between closely-related data points effectively. It can identify subtle variations that might be indicative of unique features or potential anomalies within the local context. Conversely, a larger neighborhood radius ε broadens the scope of the fuzzy neighborhood granules. This provides the model with a more comprehensive and global perspective of the data. In datasets where the data distribution is more spread-out or where there is a need to capture overarching relationships among data points, a larger ε is beneficial. However, an overly large ε can be detrimental. It may cause the fuzzy neighborhood granules to encompass an excessive number of data points, diluting the significance of individual data characteristics. This can result in a loss of important local-level information, as the model treats data points that are actually quite different as being part of the same granule. As a result, the model's ability to accurately analyze and interpret the data may be compromised, affecting its performance in anomaly detection task within the decision-free information systems. In summary, the neighborhood radius parameter ε is a key factor in the generalized FNRS model, striking a crucial balance between local-level precision and global-scale comprehensiveness, and thus significantly impacting the model's effectiveness in anomaly detection.

Definition 4. Given a decision-free $FNIS = \langle O, A, V, f, \varepsilon\rangle$, for any $o_i \in O$ and any $B \subseteq A$, we have $[o_i]_B^{\varepsilon} = \left\{[o_i]_B^{\varepsilon}(o_1), [o_i]_B^{\varepsilon}(o_2), \ldots, [o_i]_B^{\varepsilon}(o_n)\right\}$, then the fuzzy neighborhood granule structure with respect to B is defined as

$$G([o]_B^{\varepsilon}) = \left\{[o_1]_B^{\varepsilon}, [o_2]_B^{\varepsilon}, \ldots, [o_n]_B^{\varepsilon}\right\}. \tag{6}$$

The fuzzy neighborhood granule structure is a set of fuzzy neighborhood granules of all objects with respect to the same attribute subset.

As a pair of important concepts in rough set theory, upper and lower approximations are used to measure the degree of membership between different rough sets. Instead of using the fuzzy decision classification induced by D, we use the fuzzy neighborhood information granules when constructing the lower and upper approximations.

Definition 5. Given a decision-free $FNIS = \langle O, A, V, f, \varepsilon\rangle$, for any $o_i, o_k \in O$ and any $B, C \subseteq A$, $[o_i]_B^{\varepsilon}$ and $[o_k]_C^{\varepsilon}$ are the fuzzy neighborhood information granules, respectively. Then, the fuzzy neighborhood lower and upper approximations of $[o_i]_B^{\varepsilon}$ with respect to C are respectively defined as follows.

$$\underline{FN}_C^{\varepsilon}\left([o_i]_B^{\varepsilon}\right) = \left\{o_k \in O \mid [o_k]_C^{\varepsilon} \subseteq [o_i]_B^{\varepsilon}\right\}, \tag{7}$$

$$\overline{FN}_C^{\varepsilon}\left([o_i]_B^{\varepsilon}\right) = \left\{o_k \in O \mid [o_k]_C^{\varepsilon} \cap [o_i]_B^{\varepsilon} \neq \emptyset\right\}. \tag{8}$$

The lower approximation $\underline{FN}_C^{\varepsilon}\left([o_i]_B^{\varepsilon}\right)$ denotes the set of objects in O whose fuzzy neighborhood information granules with respect to C are determined to be contained in $[o_i]_B^{\varepsilon}$. And the upper approximation $\overline{FN}_C^{\varepsilon}\left([o_i]_B^{\varepsilon}\right)$ represents the set of objects in O whose fuzzy neighborhood information granules with respect to C intersect with $[o_i]_B^{\varepsilon}$. It can be seen that the upper and lower approximation is defined by establishing the relationship between different fuzzy neighborhood granules. It should be noted that typically the upper and lower approximations are a related pair of concepts, hence we define both of them for completeness. Whereas in the proposed anomaly detection method, we only use the concept of lower approximation.

Further, we define the fuzzy neighborhood lower approximation ratio for each information granule, which will be used in the subsequent definition of the intensity of anomaly.

Definition 6. Given a decision-free $FNIS = \langle O, A, V, f, \varepsilon \rangle$, for any $o_i \in O$ and any $B, C \subseteq A$, the fuzzy neighborhood lower approximation ratio of $[o_i]_B^\varepsilon$ with respect to C is denoted by

$$\rho_C^\varepsilon\left([o_i]_B^\varepsilon\right) = \frac{\left|\underline{FN}_C^\varepsilon\left([o_i]_B^\varepsilon\right)\right|}{|O|}. \tag{9}$$

For a given neighborhood radius ε and any $o_i \in O$, the lower approximation and lower approximation ratios show opposite monotonicity with respect to the attribute subsets B and C.

1) If $C_1 \subseteq C_2 \subseteq A$, then $\underline{FN}_{C_1}^\varepsilon\left([o_i]_B^\varepsilon\right) \subseteq \underline{FN}_{C_2}^\varepsilon\left([o_i]_B^\varepsilon\right)$ and $\rho_{C_1}^\varepsilon\left([o_i]_B^\varepsilon\right) \leq \rho_{C_2}^\varepsilon\left([o_i]_B^\varepsilon\right)$;
2) If $B_1 \subseteq B_2 \subseteq A$, then $\underline{FN}_C^\varepsilon\left([o_i]_{B_1}^\varepsilon\right) \supseteq \underline{FN}_C^\varepsilon\left([o_i]_{B_2}^\varepsilon\right)$ and $\rho_C^\varepsilon\left([o_i]_{B_1}^\varepsilon\right) \geq \rho_C^\varepsilon\left([o_i]_{B_2}^\varepsilon\right)$.

By now we have constructed a generalized FNRS model that can be straightforwardly applied to decision-free information systems. Next, we will discuss in detail how to implement anomaly detection based on the generalized FNRS model with the novel definition of the anomaly score.

5. Anomaly detection model based on FNRS

In this section, we first propose a method to characterize the degree of anomaly and construct the anomaly detection model based on the generalized FNRS model. Then we design an anomaly detection algorithm used for heterogeneous data and analyze its time complexity. Finally, an example of heterogeneous data is given to illustrate the algorithm procedure.

5.1. FNRS-based anomaly detection model

The decision-free $FNIS$ is expressed in a quintuple form $\langle O, A, V, f, \varepsilon \rangle$, where $O = \left\{o_1, o_2, \dots, o_n\right\}$ and $A = \left\{a_1, a_2, \dots, a_m\right\}$. To measure the outlier degree of the fuzzy neighborhood information granules, we defined the granule intensity of anomaly as follows.

Definition 7. Given a neighborhood radius ε, for any $B \subseteq A$, let $T = A - B$. For any $o_i \in O$, the granule intensity of anomaly of $[o_i]_B^\varepsilon$ is denoted by

$$GIA\left([o_i]_B^\varepsilon\right) = 1 - \frac{W_B^I\left(o_i\right) \cdot \rho_T^\varepsilon\left([o_i]_B^\varepsilon\right)}{|T| + 1}, \tag{10}$$

where $W_B^I\left(o_i\right) = \frac{\left|[o_i]_B^\varepsilon\right|}{|O|}$.

The granule intensity of anomaly is a quantitative measure of the outlier degree. For a given fuzzy neighborhood information granule $[o_i]_B^\varepsilon$, if its fuzzy neighborhood lower approximation ratio on a subset of attributes is all low, this means the information granule behaves abnormally and the $GIA\left([o_i]_B^\varepsilon\right)$ should be high. Therefore, we use the fuzzy neighborhood lower approximation ratio to define the degree of anomaly. In theory, we should compute the fuzzy neighborhood lower approximation ratio of $[o_i]_B^\varepsilon$ over all subsets of A, but this will lead to exponential increase in time complexity when the number of subsets of A is large. Therefore in Equation (10) we only choose the subset $T \subseteq A$ to construct the granule intensity of anomaly.

Definition 8. For any $o_i \in O$, the anomaly score is computed by

$$AS\left(o_i\right) = \frac{\sum_{c=1}^{m}\left(W_{\{a_c\}}^S\left(o_i\right) \cdot GIA\left([o_i]_{\{a_c\}}^\varepsilon\right)\right)}{|A|}, \tag{11}$$

where $W_{\{a_c\}}^S\left(o_i\right) = 1 - \sqrt[3]{\frac{\left|[o_i]_{\{a_c\}}^\varepsilon\right|}{|O|}}$.

The anomaly score is an indicator used to measure the probability that the object is an outlier. It is constructed by integrating the GIA and corresponding weights of the fuzzy neighborhood granules.

For any $B \subseteq A$, we can compute its granule intensity of anomaly and further use it for constructing the anomaly score. However, when the number of subsets of A is large, such computation leads to very high time complexity. Therefore, in Equation (11), we let

$B = \{a_c\}$ when computing the anomaly score. As for the design of the weight function, we consider that when an object is not very similar to other objects, it deviates more from the distribution of value domains with respect to that attribute subset, and it tends to be more likely to be an outlier of the object set. Thus we will assign it a greater weight so that its anomaly score will be higher.

For the object in the dataset, the higher the anomaly score is, the more likely the object is to be an anomaly. We can set up a score threshold to determine anomalies, thus the anomaly is defined as follows.

Definition 9. Given a score threshold η, for any $o_i \in O$, if $AS(o_i) > \eta$, then o_i is called a FNRS-based anomaly in O.

5.2. Anomaly detection algorithm

In this section, we design a specific anomaly detection algorithm, called ADFNR, based on the proposed FNRS model. We can see that the total number of loops for Algorithm 1 is $|A| \times |O| \times |O| + |A| \times |O| + |O| \times |A|$. Therefore, in the worst case, the time complexity of Algorithm 1 is $O(|A||O|^2)$.

To deal with heterogeneous data, the similarity degree between o and q on different attribute a is calculated as

$$R_a(o,q) = \begin{cases} 1 - |f(o,a) - f(q,a)|, a \text{ is numerical;} \\ 1, f(o,a) = f(q,a) \text{ and } a \text{ is nominal;} \\ 0, f(o,a) \neq f(q,a) \text{ and } a \text{ is nominal,} \end{cases} \tag{12}$$

where $f(o,a)$ and $f(q,a)$ are the attribute value of object o and q on attribute a, respectively. It should be stated that the normalization operation will be applied if the attribute a is numerical, while the nominal data will not be normalized. For similarity relations induced by nominal attributes, each element is either 0 or 1, while for similarity relations induced by numerical attributes, each element is between 0 and 1.

Algorithm 1: Anomaly detection algorithm based on fuzzy neighborhood rough set (ADFNR).

Input: Decision-free $FNIS = \langle O, A, V, f, \varepsilon \rangle$, $|O| = n$, $|A| = m$
Output: Anomaly score (AS)

$AS \leftarrow \emptyset$;
for $c \leftarrow 1$ *to* m **do**
  Calculate $G([o]^{\varepsilon}_{\{a_c\}})$;
end
for $c \leftarrow 1$ *to* m **do**
  Let $B = \{a_c\}$
  Let $T = A - B = A - \{a_c\}$
  Calculate $G([o]^{\varepsilon}_{T})$;
  for $i \leftarrow 1$ *to* n **do**
    Calculate $\rho^{\varepsilon}_{T}\left([o_i]^{\varepsilon}_{\{a_c\}}\right)$;
    Calculate $W^{I}_{\{a_c\}}(o_i)$;
    Calculate $W^{S}_{\{a_c\}}(o_i)$;
  end
end
for $i \leftarrow 1$ *to* n **do**
  for $c \leftarrow 1$ *to* m **do**
    Calculate $GIA([o_i]^{\varepsilon}_{\{a_c\}})$;
  end
  Calculate $AS(o_i)$;
end
return AS.

5.3. An illustrative example

Example 1. An $FNIS = \left\langle O = \{o_1, o_2, \ldots, o_6\}, A = \{a_1, a_2, a_3\}, V, f, \varepsilon \right\rangle$ is shown in Table 1, where a_1 is a nominal attribute and a_2, a_3 are numeric attributes.

First of all, we use the formula $a' = (a - a_{\min})/(a_{\max} - a_{\min})$ to normalize the numerical attributes a_2 and a_3. According to Equation (12), we can calculate the fuzzy similarity relation $R_{a_1}, R_{a_2}, R_{a_3}$, and further calculate the fuzzy similarity relation induced by the subset of B by formula $R_B = \cap_{a \in B} R_a$:

$$R_{a_2} = \begin{bmatrix} 1 & 0.571 & 0 & 0.143 & 0.714 & 0.143 \\ 0.571 & 1 & 0.429 & 0.571 & 0.857 & 0.571 \\ 0 & 0.429 & 1 & 0.857 & 0.286 & 0.857 \\ 0.143 & 0.571 & 0.857 & 1 & 0.429 & 1 \\ 0.714 & 0.857 & 0.286 & 0.429 & 1 & 0.429 \\ 0.143 & 0.571 & 0.857 & 1 & 0.429 & 1 \end{bmatrix}, R_{\{a_1,a_2\}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0.571 & 0.857 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0.857 \\ 0 & 0.571 & 0 & 1 & 0.429 & 0 \\ 0 & 0.857 & 0 & 0.429 & 1 & 0 \\ 0 & 0 & 0.857 & 0 & 0 & 1 \end{bmatrix}.$$

Table 1
The original (left) and normalized (right) decision-free FNIS on heterogeneous data.

O	a_1	a_2	a_3	O	a_1	a_2	a_3
o_1	B	9	0.7	o_1	B	1	1
o_2	A	6	0.3	o_2	A	0.5714	0.2
o_3	C	2	0.5	o_3	C	0	0.6
o_4	A	3	0.2	o_4	A	0.1429	0
o_5	A	7	0.4	o_5	A	0.7143	0.4
o_6	C	3	0.6	o_6	C	0.1429	0.8

According to Equation (6) with $\varepsilon = 0.65$, the fuzzy neighborhood granule structures about a single attribute or subsets of A can be obtained. The following is an example.

$G([o]^{\varepsilon}_{a_2})$ = { (1, 0, 0, 0, 0.714, 0), (0, 1, 0, 0, 0.857, 0), (0, 0, 1, 0.857, 0, 0.857), (0, 0, 0.857, 1, 0, 1), (0.714, 0.857, 0, 0, 1, 0), (0, 0, 0.857, 1, 0, 1)};

$G([o]^{\varepsilon}_{A-\{a_2\}}) = G([o]^{\varepsilon}_{\{a_1,a_3\}})$ = { (1, 0, 0, 0, 0, 0), (0, 1, 0, 0.8,0.8, 0), (0, 0, 1, 0, 0, 0.8), (0, 0.8, 0, 1, 0, 0), (0, 0.8, 0, 0, 1, 0), (0, 0, 0.8, 0, 0, 1)}.

By Definition 5, we can calculate the fuzzy neighborhood lower approximation of information granules with respect to subsets of A. Furthermore, the fuzzy neighborhood lower approximation ratio can be calculated by Definition 6, such as $\rho^{\varepsilon}_{A-\{a_1\}}\left([o_5]^{\varepsilon}_{\{a_1\}}\right) = 0.5$, $\rho^{\varepsilon}_{A-\{a_2\}}\left([o_5]^{\varepsilon}_{\{a_2\}}\right) \approx 0.1667$, $\rho^{\varepsilon}_{A-\{a_3\}}\left([o_5]^{\varepsilon}_{\{a_3\}}\right) = 0$.

According to Definition 7, we start by calculating the weights, such as $W^{I}_{\{a_1\}}\left(o_5\right) = 0.5, W^{I}_{\{a_2\}}\left(o_5\right) \approx 0.4286, W^{I}_{\{a_3\}}\left(o_5\right) \approx 0.4333$. Then for each object, we can obtain granule intensity of anomaly concerning a single attribute in A, such as $GIA\left([o_5]^{\varepsilon}_{\{a_1\}}\right) \approx 0.9167, GIA\left([o_5]^{\varepsilon}_{\{a_2\}}\right) \approx 0.9762, GIA\left([o_5]^{\varepsilon}_{\{a_3\}}\right) = 1$. In accordance with Definition 8, the weights used in Equation (11) can be computed as $W^{S}_{\{a_1\}}\left(o_5\right) = 0.2063, W^{S}_{\{a_2\}}\left(o_5\right) \approx 0.2461, W^{S}_{\{a_3\}}\left(o_5\right) \approx 0.2433$. Finally, we can obtain each object's anomaly score by integrating the granule intensity of the anomaly with corresponding weights. Taking o_5 as an example, the result is calculated as follows.

$$AS\left(o_5\right) = \frac{\sum_{c=1}^{3}\left(W^{S}_{\{a_c\}}\left(o_5\right) \cdot GIA\left([o_5]^{\varepsilon}_{\{a_c\}}\right)\right)}{|A|} = \frac{0.2061 \times 0.9167 + 0.2463 \times 0.9762 + 0.2433 \times 1.000}{3} \approx 0.224$$

In the same way, we can obtain $AS\left(o_1\right) \approx 0.369, AS\left(o_2\right) \approx 0.252, AS\left(o_3\right) \approx 0.255, AS\left(o_4\right) \approx 0.243, AS\left(o_6\right) \approx 0.251$. Therefore, object o_1 has the largest probability to be an anomaly, while object o_5 on the contrary.

6. Experiments

In this section, we conduct comparative experiments with ten anomaly detection algorithms to evaluate the feasibility and effectiveness of the proposed ADFNR algorithm. Firstly, we make some preparations for the experiment, including the datasets and the design of the experiments. The experimental results are then analyzed and parameter and statistical analyses are further carried out.

6.1. Experimental preparation

This subsection details the key preparations for our experiments. It covers three main aspects: the datasets, which form the basis for algorithm evaluation; the comparison algorithms, serving as benchmarks for the ADFNR algorithm; and the evaluation indices, used to objectively measure algorithm performance.

6.1.1. Datasets

Table 2 summarizes the basic information of twenty-five datasets used in this experiment, including the number of objects, the dimension and type of attributes, the number and percentage of anomalies. The datasets used in the experiments are all publicly available[1] and have been regularly used for anomaly detection tasks, ensuring the scientific validity and effectiveness of the experiment. The datasets accessible via this link[1] have been processed with respect to missing value filling and downsampling as mentioned in [29]. Specifically, the maximum probability method is utilized to deal with potential missing values in the dataset by substituting them with attribute values that occur with the highest frequency on other objects. Subsequently, downsampling is carried out on specific classes to create anomalies, and samples from the remaining classes are preserved. Thus, the dataset for anomaly detection is obtained.

The 25 datasets used in the experiments cover three different types, including 6 nominal, 9 numerical, and 10 heterogeneous datasets. The dataset containing both nominal and numerical attributes is called a heterogeneous dataset. As can be seen from

[1] https://github.com/BELLoney/Outlier-detection.

Table 2
Description of twenty-five datasets.

No.	Datasets	Abbreviation	Type	# Object	# Dimension	# Anomaly	% Anomaly
1	Arrhythmia_variant1	Arrhythmia	Heterogeneous	452	279	66	14.60
2	Bands_band_34_variant1	Bands	Heterogeneous	346	39	34	9.83
3	Cardiotocography_2and3_33_variant1	Cardiotocography	Numerical	1688	21	33	1.95
4	Chess_nowin_185_variant1	Chess185	Nominal	1854	36	185	9.98
5	Chess_nowin_227_variant1	Chess227	Nominal	1896	36	227	11.97
6	CreditA_plus_42_variant1	Credit	Heterogeneous	425	15	42	9.88
7	Diabetes_tested_positive_26_variant1	Diabetes	Numerical	526	8	26	4.94
8	German_1_14_variant1	German	Heterogeneous	714	20	14	1.96
9	Heart270_2_16_variant1	Heart	Heterogeneous	166	13	16	9.64
10	Hepatitis_2_9_variant1	Hepatitis	Heterogeneous	94	19	9	9.57
11	Horse_1_12_variant1	Horse	Heterogeneous	256	27	12	4.69
12	Ionosphere_b_24_variant1	Ionosphere	Numerical	249	34	24	9.64
13	Iris_Irisvirginica_11_variant1	Iris	Numerical	111	4	11	9.91
14	Lymphography	Lymphography	Nominal	148	18	6	4.05
15	Pageblocks_1_258_variant1	Pageblocks	Numerical	5171	10	258	4.99
16	Pima_TRUE_55_variant1	Pima	Numerical	555	9	55	9.91
17	Sick_sick_35_variant1	Sick35	Heterogeneous	3576	29	35	0.98
18	Sick_sick_72_variant1	Sick72	Heterogeneous	3613	29	72	1.99
19	Sonar_M_10_variant1	Sonar	Numerical	107	60	10	9.35
20	Thyroid_disease_variant1	Thyroid	Heterogeneous	9172	28	74	0.81
21	Tic_tac_toe_negative_26_variant1	Tic26	Nominal	652	9	26	3.99
22	Tic_tac_toe_negative_32_variant1	Tic32	Nominal	658	9	32	4.86
23	Vote_republican_29_variant1	Vote	Nominal	296	16	29	9.80
24	Wbc_malignant_39_variant1	Wbc	Numerical	483	9	39	8.07
25	Yeast_ERL_5_variant1	Yeast	Numerical	1141	8	5	0.44

Table 2, the number of objects ranged from 94 to 9172 with attribute dimensions from 4 to 279. The percentage of anomalies is distributed in the interval of [0.44%, 14.60%]. All experiments were conducted on the device equipped with a 12th Gen Intel Core i9-12900K processor and 128 GB RAM, with the operating system Windows 11. The development environment is PyCharm, using the Python programming language.

6.1.2. Algorithms for comparison

To verify the effectiveness of the proposed ADFNR algorithm, it is compared with ten outlier detection algorithms, which are K-Nearest Neighbors (KNN) [30], Kernel Density Estimation (KDE) [31], Isolation Forest (IForest) [32], Natural Outlier Factor-based method (NOF) [33], Local Projection-based Outlier Detection (LPOD) [34], COPula-based Outlier Detector (COPOD) [35], Directed density ratio Changing Rate-based Outlier Detection (DCROD) [36], Empirical Cumulative-based Outlier Detection (ECOD) [37], Incomplete Local and Global Neighborhood Information (ILGNI) network [38], Weighted Fuzzy-Rough Density-based Anomaly detection (WFRDA) [39].

Among them, KNN is a distance-based algorithm that classifies or predicts by leveraging the classes or values of a data point's K nearest neighbors, and KDE is an approach for estimating probability density functions by applying kernel functions to data points, ECOD and COPOD are based on empirical cumulative distribution functions, DCROD and WFRDA are density-based algorithms, ILGNI is a graph-based detection method. Fuzzy rough sets are used to define fuzzy-rough densities in WFRDA, which consider both density and uncertainty information of samples [39]. Based on fuzzy rough sets, FNRS further considers the neighborhood information of samples. Therefore, in this paper, we designed the ADFNR algorithm based on the proposed generalized FNRS. In our experiments, the number of base estimators of IForest is set to 100 as it is a commonly utilized default value in comparable research. Additionally, the relevant study [32] also reveals that the path lengths typically converge satisfactorily before reaching this value, signifying that setting it to 100 can offer relatively stable and efficient performance for our experimental configuration. In KDE, the bandwidth is automatically calculated using Silverman's rule, which helps determine the spread of the kernel function and is crucial for accurately estimating and smoothing the probability density function. KKN, LPOD and DCROD mainly involve the parameter k, thus we vary k from 1 to 60 with step size 1 to find the optimal parameter setting. We adjust the parameter σ of WFRDA in [0.1, 2] with a step size of 0.1. For ADFNR, we set the parameter adjustment range to [0, 1] with a step size of 0.1. The optimal parameter settings of KNN, LPOD, DCROD, WFRDA, and ADFNR for different datasets are shown in Table 3.

6.1.3. Evaluation index

To evaluate the performance of the algorithm, the Receiver Operating Characteristic (ROC) curve and Area Under Curve (AUC) are used in the experiments, both of which have been widely used in anomaly detection studies [40].

Given a score threshold η, objects with AS greater than the threshold are considered outliers, and the set of predicted outliers O_P is finally obtained as follows:

$$O_P = \left\{ o_i | AS(o_i) > \eta \right\}. \tag{13}$$

Let O_R represent the set of real outliers in O. By comparing the similarities and differences between the O_R and O_P, the True Positive Rate (TPR) and the False Positive Rate (FPR) are respectively computed by

Table 3
Optimal parameter settings for different datasets.

Datasets	k (KNN)	k (LPOD)	k (DCROD)	σ (WFRDA)	ε (ADFNR)
Arrhythmia	60	44	59	0.5	0.7
Bands	60	60	48	0.6	0.55
Cardiotocography	59	5	60	0.8	0.5
Chess185	46	16	22	0.1	0.1
Chess227	43	15	25	0.1	0.1
Credit	60	36	59	0.1	0.9
Diabetes	8	27	59	0.3	0.9
German	24	57	14	0.5	0.9
Heart	45	7	60	0.2	0.8
Hepatitis	49	39	60	0.5	0.55
Horse	49	56	60	0.1	1
Ionosphere	1	49	3	0.5	0.75
Iris	45	4	50	0.2	0.2
Lymphography	36	9	59	0.1	0.1
Pageblocks	60	22	60	2	0.5
Pima	12	21	60	0.3	0.9
Sick35	2	5	14	0.1	1
Sick72	4	60	14	0.1	1
Sonar	28	10	45	0.3	0.75
Thyroid	60	46	60	0.4	1
Tic26	10	30	5	0.1	0.1
Tic32	9	30	4	0.1	0.1
Vote	60	60	3	0.1	0.1
Wbc	60	10	60	1	0.2
Yeast	60	2	47	1	0.25

$$TPR = \frac{|O_P \cap O_R|}{|O_R|}, \tag{14}$$

$$FPR = \frac{|O_P - O_R|}{|O - O_R|}. \tag{15}$$

The ROC curve is plotted with FPR on the x-axis and TPR on the y-axis. In general, for an effective anomaly detection model, it is desired that the FPR be as small as possible while the TPR be as large as possible. Therefore, for each algorithm, the closer the ROC curve is to the upper left corner, the better its performance will be. The ROC curve is intuitive, but it is difficult to measure some of the subtle differences, especially when two curves cross. So we further use the AUC value to make more precise evaluations of the algorithms. The AUC value is defined as the area enclosed by the axes under the ROC curve, which ranges from 0 to 1. For the same dataset, algorithms with higher AUC values have better detection performance.

6.2. Experimental results analysis

In this subsection, we systematically analyze the experimental results. Our analysis encompasses multiple vital aspects. First, we start with the performance on the ROC curve, which is a fundamental metric for evaluating the performance of anomaly-detection algorithms. Then, we'll move on to the performance on AUC, a comprehensive measure that reflects the overall effectiveness of the algorithms. Additionally, we conduct a parameter analysis to explore how the parameter affects the performance of the proposed algorithm. Finally, a statistical analysis will be performed to evaluate the statistical performance of each algorithm on the AUC index.

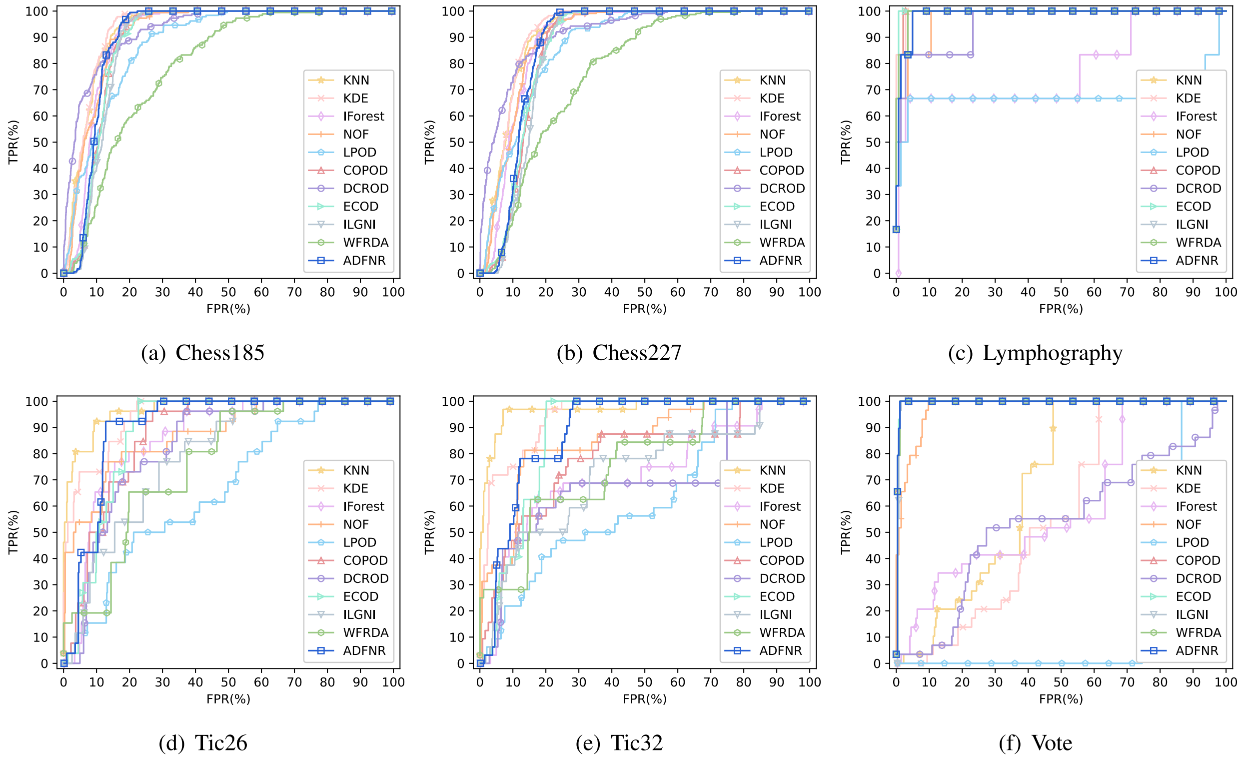
6.2.1. Performance on ROC and AUC

Figs. 1–3 show the ROC curves for each of the eleven anomaly detection algorithms on the nominal, numerical, and heterogeneous datasets, respectively. It can be seen that ADFNR attains the best performance among eleven detection algorithms on most datasets. On Vote, Diabetes, Iris, Pima, Wbc, Yeast, Credit, and Horse datasets, the ROC curves of ADFNR are almost above the curves of other algorithms. In particular, on the Iris and Yeast datasets, the ROC curves of ADFNR passed the optimal point.

When ROC curves overlap or cross, it is difficult to determine which algorithm is better. Therefore, we use AUC as another evaluation index to more accurately assess the performance of the algorithms. Table 4 shows the AUC values of the 11 algorithms on 25 datasets, where the best results are **bolded**, and the second best results are <u>underlined</u>. As can be seen, the average AUC values for KNN, KDE, IForest, NOF, LPOD, COPOD, DCROD, ECOD, ILGNI, WFRDA, and ADFNR are 0.913, 0.903, 0.885, 0.915, 0.830, 0.913, 0.879, 0.920, 0.898, 0.902, 0.934, respectively, where ADFNR has the largest mean AUC value.

ADFNR obtained the optimum on a total of 13 datasets. In addition, ADFNR achieved the second best on 17 datasets, and achieved the fourth best on 22 datasets except for Chess185, Chess227, and Lymphography datasets. In particular, the AUC value reaches 1.000 on the Iris, and Yeast datasets. From Table 4, we can find that the algorithm ADFNR achieved optimal results on nominal, numerical and heterogeneous datasets, which illustrates that the proposed algorithm is effective in dealing with different types of datasets.

Table 5 illustrates the average ranks of eleven algorithms, and the number of datasets in which they achieved top-1 AUC, top-2 AUC, and top-4 AUC. As can be seen, ADFNR demonstrates an obvious advantage in all four indices. In particular of the top-1 AUC,



(a) Chess185 (b) Chess227 (c) Lymphography

(d) Tic26 (e) Tic32 (f) Vote

Fig. 1. Comparison of ROC curves on nominal datasets.

ADFNR achieves the best performance with 13 top-1 AUCs, followed by KNN, KDE, LPOD and WFRDA with 5, and COPOD with 3. For the amount of top-2 AUC, ADFNR is still obviously superior against the other ten algorithms, which have less than 10 top-2 AUCs. On 22 datasets, ADFNR received the top-4 AUC. It can be seen the average ranks for KNN, KDE, IForest, NOF, LPOD, COPOD, DCROD, ECOD, ILGNI, WFRDA, and ADFNR are 4.6, 5.48, 6.72, 5.44, 8.12, 5.6, 7.56, 5.28, 7.56, 5.2, 2.32, respectively, where ADFNR achieved the highest average rank.

6.3. Parameter analysis

The parameter ε plays an important role in ADFNR, which affects the anomaly score by controlling the neighborhood radius of the fuzzy neighborhood granules. Therefore, we conduct a sensitivity analysis on the parameter ε in this subsection. Given that the fuzzy similarity varies in [0,1], the neighborhood radius ε is set to increase sequentially from 0 and 1 with steps of 0.05. As can be seen from Equation (12), a variation in parameter ε does not change the similarity between the nominal attributes. Therefore, we perform the parameter analysis only on the numerical and heterogeneous datasets.

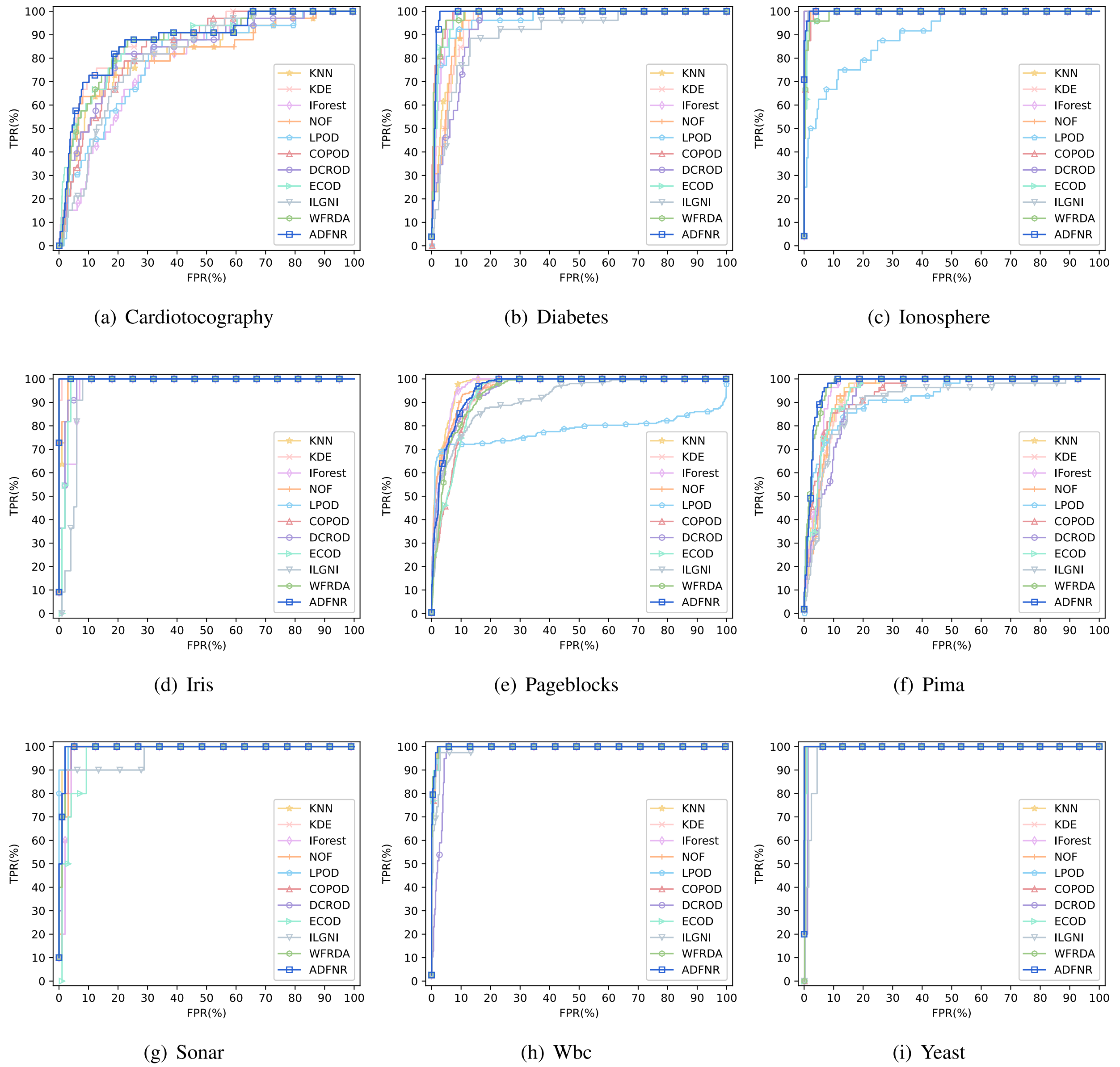
Fig. 4 shows the AUC values of the proposed ADFNR with parameter ε taking different values on numerical and heterogeneous datasets. We can find that there are small fluctuations in AUC on most datasets with ε varying between 0.3 and 0.9. When the ε is greater than 0.9, most of the curves show a decreasing trend, especially the curve of the Ionosphere dataset. The AUC declines because when the radius is larger than most similarities, the fuzzy similarity matrix becomes sparse, resulting in the loss of some important information. However, the results of Horse, Sick35, and Sick72 datasets are reversed, whose AUC values are still on an increasing trend when ε becomes greater. For these datasets, the larger neighborhood radius may allow them to discard redundant information, making it easier to detect anomalies.

6.4. Statistical analysis

We use Friedman's test [41] and Nemenyi's test [42] to evaluate the statistical performance of each algorithm on the AUC index. At first, the AUC value of each algorithm on all datasets is sorted in ascending order to obtain a specific rank. Then, Friedman's test is calculated as follows.

$$\tau_{\chi^2} = \frac{12N}{h(h+1)}\left(\sum_{i=1}^{h} R_i^2 - \frac{h(h+1)^2}{4}\right), \tag{16}$$

$$\tau_F = \frac{(N-1)\tau_{\chi^2}}{N(h-1) - \tau_{\chi^2}}, \tag{17}$$



(a) Cardiotocography (b) Diabetes (c) Ionosphere (d) Iris (e) Pageblocks (f) Pima (g) Sonar (h) Wbc (i) Yeast

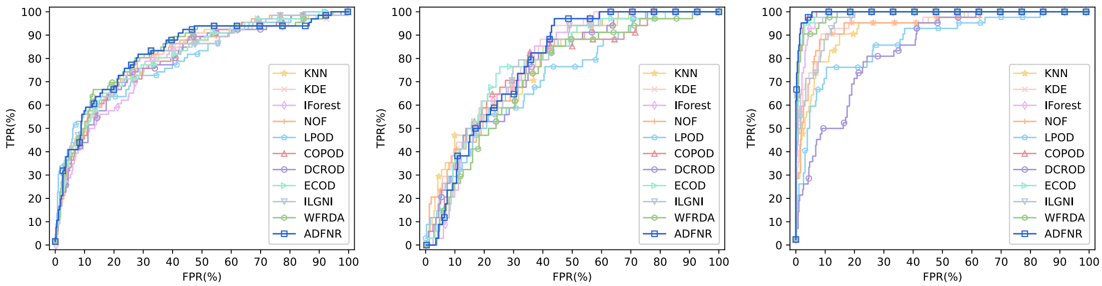
Fig. 2. Comparison of ROC curves on numerical datasets.

where N and h represent the number of datasets and algorithms, and R_i denotes the average rank of the ith algorithm over all datasets.

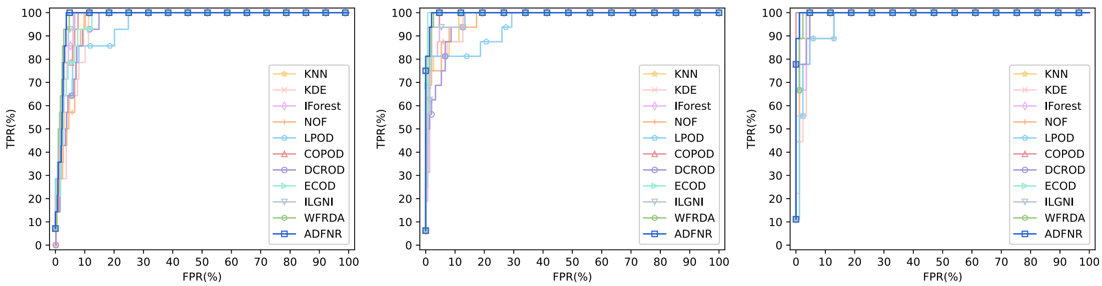
Friedman's test serves as a statistical approach that can effectively discern whether a significant statistical difference exists among the performances of h algorithms. However, its limitation lies in the fact that while it can establish the presence of a performance variance among at least two algorithms, it falls short in precisely identifying which specific pairs of algorithms exhibit these differences. In light of this, Nemenyi's post-hoc test is strategically employed to conduct a more in-depth analysis of the differences between particular pairs of algorithms. The Critical Difference (CD) among these anomaly detection algorithms is calculated by

$$CD_\alpha = q_\alpha \sqrt{\frac{h(h+1)}{6N}}, \tag{18}$$

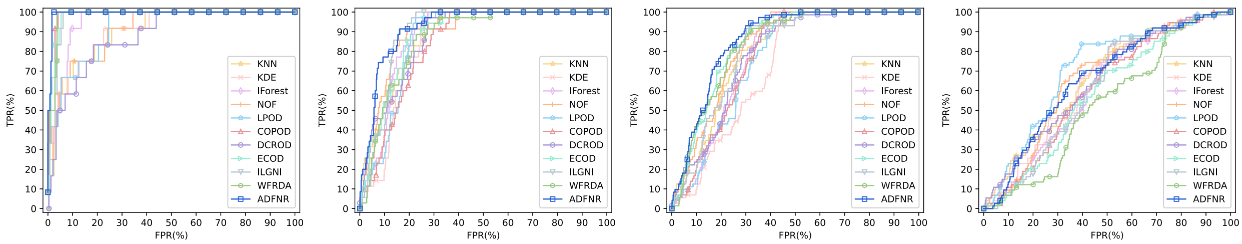
where q_α is the critical value that can be obtained by looking up the table with the specified number of algorithms. Following [43], a CD diagram is drawn to visually display the significant differences among all algorithms. Each dot on the horizontal axis represents the average rank of the algorithm in descending order of AUC. For any two algorithms, if the difference in their average rank is smaller than the CD value, they will be connected by a horizontal line segment, which means that there is no significant difference between them.



(a) Arrhythmia (b) Bands (c) Credit

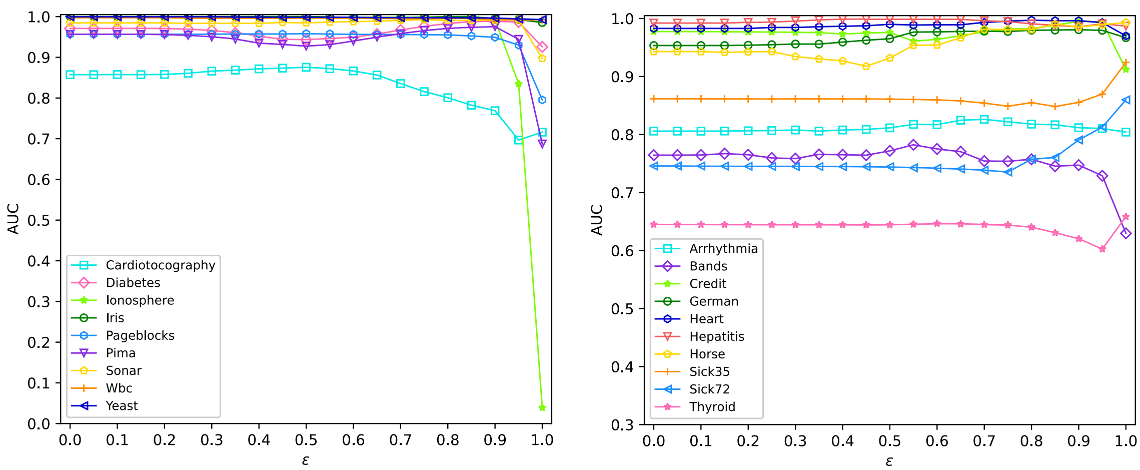


(d) German (e) Heart (f) Hepatitis



(g) Horse (h) Sick35 (i) Sick72 (j) Thyroid

Fig. 3. Comparison of ROC curves on heterogeneous datasets.



(a) Numerical (b) Heterogeneous

Fig. 4. The variation curve of AUC on parameter ε.

Table 4
Comparison of AUC values of eleven algorithms on twenty-five datasets.

Dataset	KNN	KDE	IForest	NOF	LPOD	COPOD	DCROD	ECOD	ILGNI	WFRDA	ADFNR
Arrhythmia	0.808	0.806	0.788	0.816	0.799	0.805	0.796	0.807	0.816	**0.826**	**0.826**
Bands	0.778	**0.800**	0.776	0.776	0.722	0.763	0.762	0.782	0.775	0.729	0.782
Cardiotocography	0.818	0.875	0.788	0.817	0.791	0.844	0.834	0.871	0.811	0.865	**0.876**
Chess185	0.929	**0.932**	0.907	0.916	0.878	0.890	0.929	0.890	0.883	0.789	0.901
Chess227	0.915	0.917	0.894	0.901	0.871	0.865	**0.919**	0.865	0.856	0.772	0.876
Credit	0.930	0.946	0.983	0.941	0.873	0.992	0.841	0.990	0.964	0.986	**0.995**
Diabetes	0.958	0.952	0.976	0.954	0.966	0.986	0.935	0.979	0.905	0.984	**0.988**
German	0.960	0.950	0.975	0.955	0.949	0.973	0.955	0.966	0.980	**0.984**	0.981
Heart	0.977	0.976	0.984	0.974	0.949	0.993	0.970	0.996	0.983	0.995	**0.997**
Hepatitis	0.991	0.976	0.988	0.975	0.970	**1.000**	0.990	0.996	0.995	0.995	0.999
Horse	0.902	0.901	0.954	0.908	0.911	0.986	0.878	0.981	0.975	0.982	**0.993**
Ionosphere	**1.000**	**1.000**	0.999	0.999	0.900	0.995	**1.000**	0.994	0.998	0.993	0.999
Iris	0.992	0.999	0.971	0.994	**1.000**	**1.000**	0.983	0.977	0.951	**1.000**	**1.000**
Lymphography	0.994	0.995	0.781	0.977	0.673	0.994	0.954	**0.996**	0.991	0.993	0.987
Pageblocks	**0.973**	0.956	0.970	0.963	0.779	0.939	0.953	0.938	0.914	0.944	0.958
Pima	0.943	0.939	0.957	0.942	0.920	0.942	0.928	0.947	0.901	0.974	**0.975**
Sick35	0.882	0.852	0.881	0.894	0.867	0.839	0.875	0.883	0.902	0.868	**0.924**
Sick72	0.804	0.728	0.784	0.826	0.766	0.780	0.788	0.843	0.807	0.837	**0.860**
Sonar	0.992	0.979	0.976	0.985	**0.997**	0.986	0.989	0.966	0.965	0.991	0.993
Thyroid	0.660	0.637	0.614	0.662	**0.708**	0.611	0.646	0.581	0.610	0.531	0.659
Tic26	**0.969**	0.946	0.862	0.875	0.666	0.872	0.826	0.881	0.789	0.764	0.904
Tic32	**0.971**	0.942	0.721	0.853	0.622	0.785	0.723	0.878	0.722	0.758	0.886
Vote	0.681	0.569	0.603	0.976	0.166	**0.995**	0.537	**0.995**	0.994	**0.995**	**0.995**
Wbc	**0.997**	**0.997**	0.996	**0.997**	**0.997**	0.995	0.976	0.995	0.988	**0.997**	**0.997**
Yeast	0.998	**1.000**	0.997	**1.000**	**1.000**	0.997	0.990	0.995	0.981	0.998	**1.000**
Average	0.913	0.903	0.885	0.915	0.830	0.913	0.879	0.920	0.898	0.902	**0.934**

Table 5
Number of datasets for eleven algorithms reaching top-x AUC and their average ranks.

AUC	KNN	KDE	IForest	NOF	LPOD	COPOD	DCROD	ECOD	ILGNI	WFRDA	ADFNR
Top-1	5	5	0	2	5	3	2	2	0	5	**13**
Top-2	6	10	1	3	5	6	3	5	1	6	**17**
Top-4	11	10	4	10	5	8	3	14	4	14	**22**
Avg_Rank	4.60	5.48	6.72	5.44	8.12	5.60	7.56	5.28	7.56	5.20	**2.32**

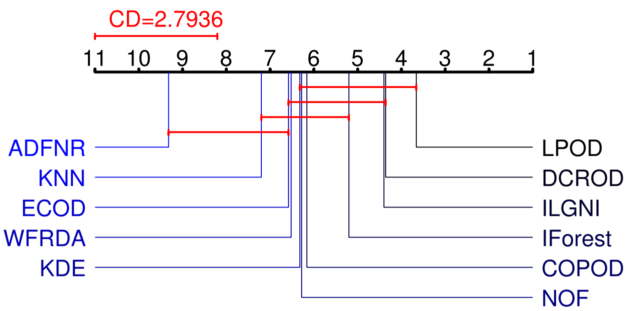


Fig. 5. Nemenyi's test.

Specifically, we employ the statistical analysis to a total of 11 algorithms on 25 datasets. With significance level $\alpha = 0.1$, we get the value of $\tau_F = 6.9281$, which is greater than the critical value. So we reject the original hypothesis that all compared algorithms perform equally. Furthermore, Nemenyi's test diagram with $CD = 2.7936$ and significance level $\alpha = 0.1$ is shown in Fig. 5. It can be seen that ADFNR was not connected with WFRDA, KDE, COPOD, LPOD, DCROD, ILGNI, IForest, and NOF. Therefore, there are significant differences between ADFNR and the comparison algorithms except for KNN and ECOD. And ADFNR obtains the best performance compared with the other 10 algorithms.

7. Conclusions and future works

In this study, we put forward a generalized FNRS model, successfully extending the FNRS theory to decision-free information systems. Based on this model, an anomaly detection method for heterogeneous data, along with the ADFNR algorithm, was developed. The ADFNR algorithm, by introducing parametric fuzzy relations, effectively copes with different data types and combines the strengths of FNRS theory and parameterized fuzzy relations to handle heterogeneous and fuzzy data. Our experimental results on 25 datasets, comparing ADFNR with 10 other algorithms, clearly demonstrate its superiority. ADFNR achieved the highest average AUC value of 0.934, outperforming the others. It obtained the optimum on 13 datasets, and ranked second on 17 datasets and forth on 22 datasets. Across different data types, ADFNR showed remarkable effectiveness, achieving optimal results on nominal, numerical and heterogeneous datasets. In terms of average ranks, ADFNR ranked highest with a value of 2.32, and also had clear advantages in the number of datasets achieving top-1, top-2, and top-5 AUC. Statistical tests further validated the significance of ADFNR's performance. With a significance level of $\alpha = 0.1$, the value of $\tau_F = 6.9281$ was greater than the critical value, leading to the rejection of the null hypothesis that all compared algorithms perform equally. The Nemenyi's test revealed significant differences between ADFNR and most comparison algorithms, except KNN and ECOD. In conclusion, the proposed ADFNR algorithm has proven to be highly effective, applicable, and stable in anomaly detection for various data types.

From the point of view of the application and the model itself, future developments will consider two aspects: practical application scenarios and automatic determination of the optimal parameters. For one thing, the application of the proposed model to realistic scenarios, such as health insurance fraud detection, is considered. It remains a challenge to efficiently and accurately detect fraud from complex, unbalanced, and inconsistent health insurance data. For another, a fuzzy granulation-based neural network approach is considered to design an optimization algorithm to enable it to find the optimal parameters automatically.

CRediT authorship contribution statement

Yuan Yuan: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Sihan Wang:** Writing – review & editing, Visualization, Investigation, Formal analysis. **Hongmei Chen:** Supervision, Project administration, Funding acquisition. **Chuan Luo:** Supervision, Project administration, Funding acquisition. **Zhong Yuan:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data and code are publicly available online at https://github.com/yuyuan97/ADFNR.

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